

Automated Reverse Port Scanning Lab

Complete Windows commands and demonstration steps

1. Open the project folder

```
cd C:\Users\student\Downloads\Automated_Reverse_Port_Scanning_Lab
dir
```

2. Check Python

```
python --version
```

3. Fix requirements for Python 3.7

```
notepad requirements.txt
```

Replace requirements.txt with

```
Flask==2.2.5
```

4. Install Flask

```
python -m pip install Flask==2.2.5
python -c "import flask; print(flask.__version__)"
```

5. Check configuration

```
type config.json
```

6. Start the honeypot and dashboard

```
python app.py
```

7. Open the dashboard

```
http://127.0.0.1:5000
```

8. Test the honeypot in a second PowerShell/CMD window

```
cd C:\Users\student\Downloads\Automated_Reverse_Port_Scanning_Lab
python -c "import socket; s=socket.create_connection(('127.0.0.1',8080)); print('CONNECTED'); print(s.recv(1024))"
```

9. Verify port 8080

```
netstat -ano | findstr :8080
```

10. Check the attack database

```
python -c "import sqlite3; c=sqlite3.connect('attacks.db'); print(c.execute('SELECT * FROM events').fetchall())"
```

11. Show latest events

```
python -c "import sqlite3; c=sqlite3.connect('attacks.db'); print(c.execute('SELECT id,source_ip,scan_status FROM events ORDER BY id DESC LIMIT 5').fetchall())"
```

12. Run the authorized scanner manually

```
python scanner.py --target 127.0.0.1
```

13. Full demonstration flow

```
WINDOW 1
cd C:\Users\student\Downloads\Automated_Reverse_Port_Scanning_Lab
python app.py
```

```
WINDOW 2
```

```
cd C:\Users\student\Downloads\Automated_Reverse_Port_Scanning_Lab
python -c "import socket; s=socket.create_connection(('127.0.0.1',8080)); print('CONNECTED'); print(s.recv(1024))"

BROWSER
http://127.0.0.1:5000

DATABASE
python -c "import sqlite3; c=sqlite3.connect('attacks.db'); print(c.execute('SELECT id,source_ip,scan_status').fetchall())"
```

14. What to show your teacher

1. Start app.py.
2. Show TCP 8080 listening.
3. Make a test connection.
4. Show source IP detection.
5. Show the SQLite event.
6. Show the completed authorized reverse scan.
7. Show discovered open ports.
8. Open the Flask dashboard.

15. Safety

Use this scanner only against your own computer/VMs or systems for which you have explicit authorization. The project checks config.json before scanning a target.